FACULTY OF COMPUTING
UTM Johor Bahru

SECR 1013
DIGITAL LOGIC

NETWORK PACKET TRANSMISSION MONITORING SYSTEM

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DEDICATION AND ACKNOWLEDGEMENT

First and foremost, we want to express our gratitude to Sir Ahmad Fariz Bin Ali, our lecturer, who has always supported us and done his utmost to ensure that we finished this project flawlessly. His passion for teaching us Digital Logic subjects truly helps our comprehension in understanding of the task given.

Moreover, the project has been successfully done undoubtedly thanks to each team member who had given the best cooperation in completing the task given to them. Without the effort and trust that had been poured in, it would not be possible to submit this project on time. We are more than grateful that all of us became teammates with a lot of creative ideas and possible discussions while doing the circuit.

Not to forget, a big thanks to every other person who was involved in this project intentionally or unintentionally. We acknowledge all the help that is given in order to make sure this project is completely done.

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1.0 INTRODUCTION

This project aims to implement the knowledge acquired in class by designing and simulating a real-world packet transmission system using Deeds Software. The project involves transferring packets from the source computer in Lab A to a destination computer in Lab B through a single cable connection. The system utilizes MUX, DEMUX, counters, comparators, clock enablers, and monitoring modules to ensure efficient data transfer.

2.0 PROBLEM BACKGROUND

In a networked computing environment, data transmission between multiple devices must be both accurate and efficient. Without a well-structured system, packet loss, congestion, or inefficient routing could disrupt communication, leading to unreliable data exchange. In this project, we examine the challenge of transmitting packets from one lab to another using a single cable while ensuring proper monitoring and control over the data flow.

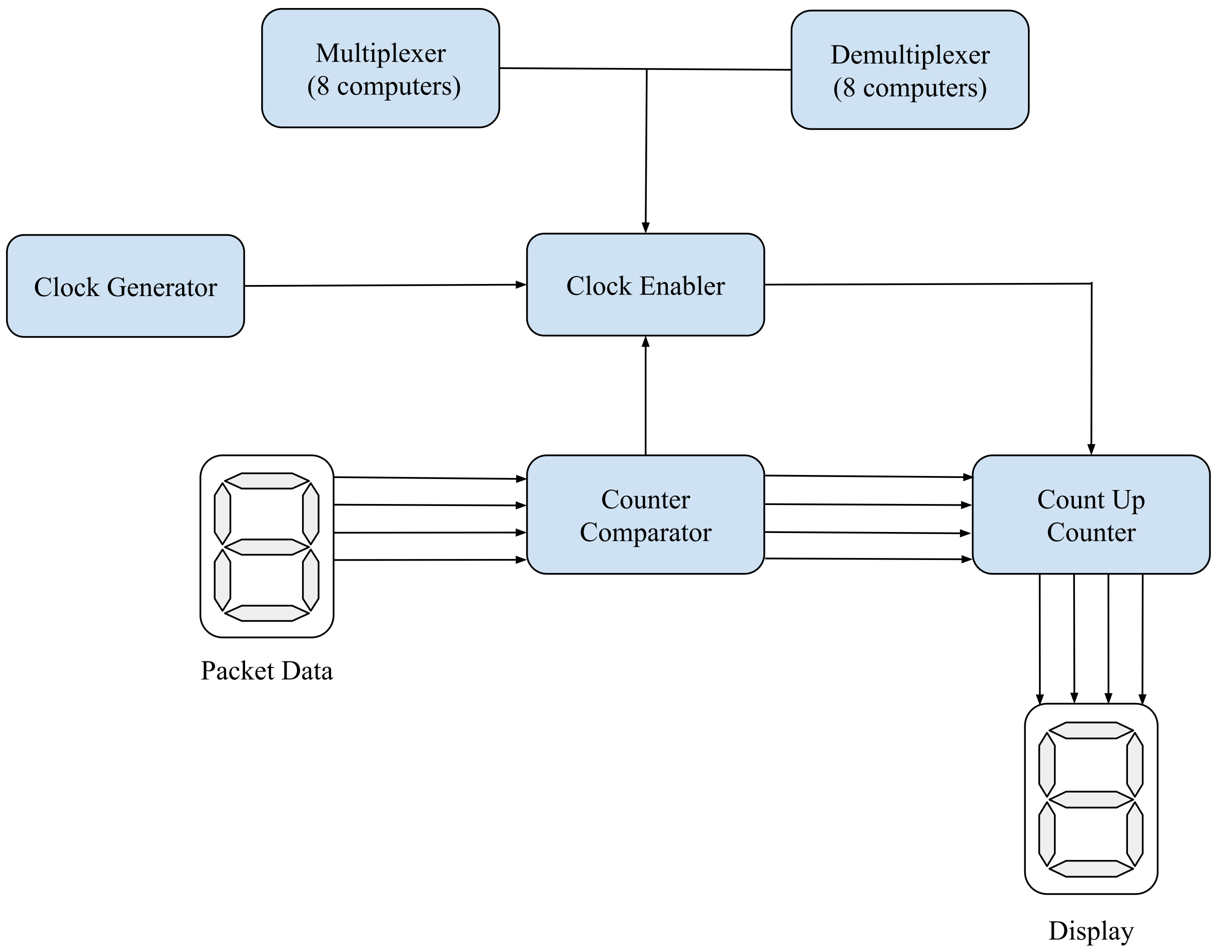
A major issue in network communication is managing multiple devices efficiently when they share a common transmission medium. The use of multiplexers and demultiplexers ensures that packets reach the correct destination without interference. Another crucial aspect is tracking the number of packets transmitted to prevent errors or data loss. By implementing a synchronous counter T flip-flops, we can accurately monitor the packet count and stop transmission when the desired limit is reached.

Additionally, clock enablers help regulate the timing and synchronization of data transfer, ensuring that the process runs smoothly. A well-structured monitoring system provides real-time updates on transmission progress, which is useful for diagnosing potential issues. The addition of a fun display feature in the advanced implementation adds an engaging visual representation of the packet transfer process, making the system more interactive.

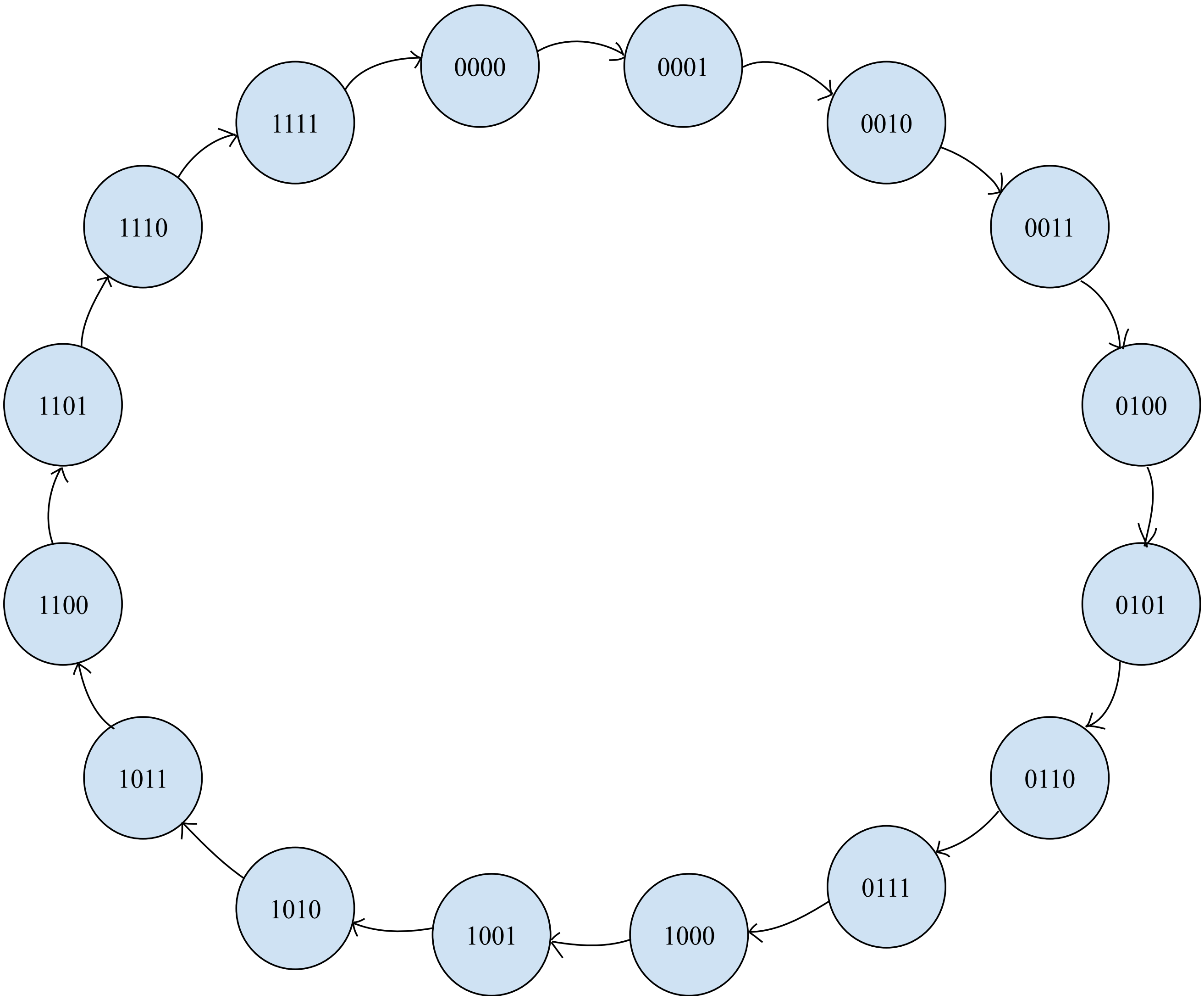
3.0 SUGGESTED SOLUTION

The block diagram of the required components is shown in Figure 3.1. Firstly, turn on the computers in Lab A which is the source and Lab B the destination that will be used for packet transmission. The LED will light up only if both the sender and receiver computers are powered on. Then, the clock enabler must be activated to allow data transmission. When enabled, it allows the counter to start counting packet transfers. The blinking LED indicates that the clock is in toggling mode, confirming that the system is actively processing the transmission. The clock enabler is connected to the counter circuit, ensuring that packet counting starts only when the clock is operational. A T flip-flops-based synchronous counter is used to track the number of packets being transmitted. The counter increments with each transmitted packet and serves as an input for the comparator. The flip-flops are also connected to the seven-segment display which is DispHex to visually represent the number of packets sent. The PRE(Preset) and CLR(Clear) inputs of the flip-flop are used to initialize and reset the counter when needed. Then, the user selects the packet data to be transmitted. The selected data is sent through a multiplexer (MUX) in Lab A and then transmitted via a single cable. At Lab B, a demultiplexer (DEMUX) routes the received data to the correct destination computer. The comparator continuously compares the current packet count with the maximum packet limit set by the user. If the count matches the maximum value, the system stops transmission, preventing unnecessary data flow. The packet data is displayed in DispHex, allowing users to monitor the ongoing transmission. By following this design, the packet transmission process is controlled efficiently, ensuring reliable communication between the computers in Lab A and Lab B.

3.1 Block Diagram



3.2 State Diagram



3.3 Next Step and Flip-Flop Transition Table

Input, X	Present State				Next State				T Flip-Flop			
	$Q3$	$Q2$	$Q1$	$Q0$	$Q3_+$	$Q2_+$	$Q1_+$	$Q0_+$	$T3$	$T2$	$T1$	$T0$
0	0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	1	0	0	1	0	0	0	1	1
0	0	0	1	0	0	0	1	1	0	0	0	1
0	0	0	1	1	0	1	0	0	0	1	1	1
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1	1	0	1	0	1	0	1	1	0	0	0	1
1	1	0	1	1	1	1	0	0	0	1	1	1
1	1	1	0	0	1	1	0	1	0	0	0	1
1	1	1	0	1	1	1	1	0	0	0	1	1
1	1	1	1	0	1	1	1	1	0	0	0	1
1	1	1	1	1	0	0	0	0	1	1	1	1

3.4 Karnaugh Map (K-Map)

Q_0Q_1 \ Q_2Q_3	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	0	1	0
10	0	0	0	0

$$T_0 = Q_1Q_2Q_3$$

$$T_1 = Q_2Q_3$$

Q_0Q_1 \ Q_2Q_3	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	0	1	0
10	0	0	1	0

Q_0Q_1 \ Q_2Q_3	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

$$T_2 = Q_3$$

$$T_3 = 1$$

Q_0Q_1 \ Q_2Q_3	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

4.0 THE REQUIREMENT

For the implementation of this project, the following requirements are used:

4.1 Software Requirement

Deeds (Digital Circuit Simulation Software)

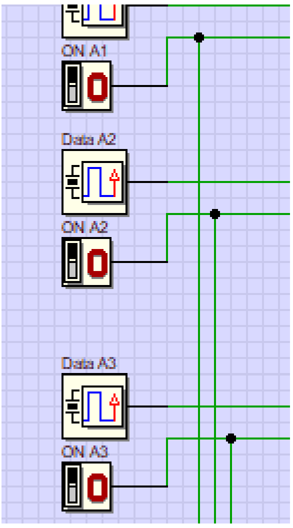
4.2 Logical Components

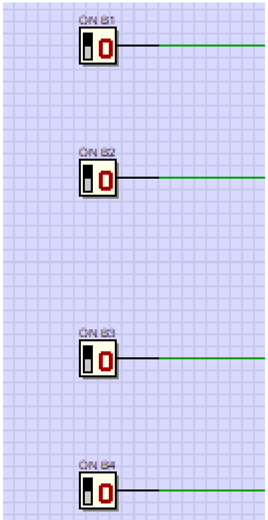
1. Multiplexer (MUX)
2. Demultiplexer (DEMUX)
3. AND gates
4. OR gates
5. NOT gates
6. Counters (T flip-flop)
7. Comparator
8. Clock generator
9. Test LED
10. Input switch
11. Input hex digit (packet data input)
12. Output one bit
13. Seven-segment display (one hex digit)
14. Wires (connections)

5.0 SYSTEM IMPLEMENTATION

5.1 Input Switch

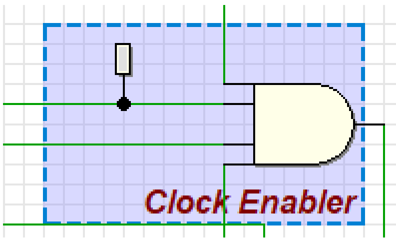
The usage of input switches allows users to interact with the circuit and enter any relevant input. There are 16 input switches required to turn on the computers in Lab A and Lab B. Turning the switch ON represents that the computer is active and ready for packet transmission. If both the sender and receiver computers are ON, the LED indicator will light up, signaling that transmission can proceed. Other than that, some input switches are used to select the packet data to be transmitted. The clock enabler also needs an input switch activation to count packet transfers. A few input switches also control the reset and preset operations of the counter using the CLR (Clear) and PRE (Preset) functions on the flip-flop.





5.2 Clock Enabler

The clock enabler controls when the counter operates. This component uses AND gates to check multiple conditions such as power ON, packet transmission mode, and comparator output before allowing the clock to function. It also ensures the counter only works when needed, preventing unnecessary counting.

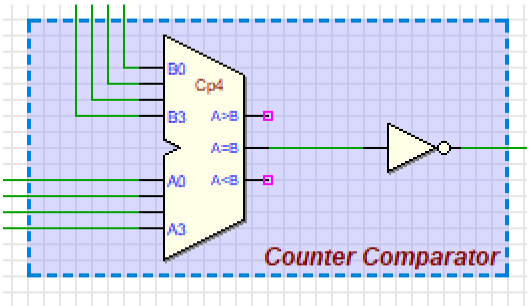


5.3 Multiplexer (MUX) & Demultiplexer (DEMUX)

We used MUX which allows one of the 8 computers in Lab A to send data through a single cable. The DEMUX in Lab B receives the data and directs it to the correct destination computer. Moreover, logical connections use OR gates to combine multiple computer signals and control selection inputs for data transmission.

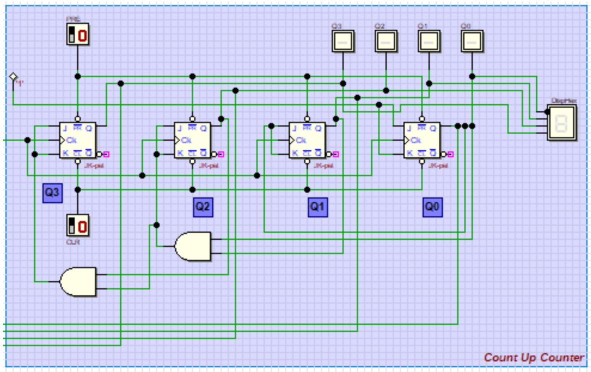
5.4 Comparator

This component compares the current packet count with the maximum packet count set by the user. If the counts do not match, the system continues transmitting and once the counts are equal, the comparator stops the counter to stop transmission.



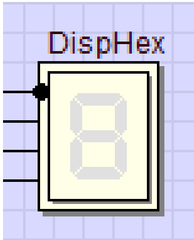
5.5 Counter (T Flip-Flop)

We implement the counter to monitor the number of packets transmitted. A 4-bit synchronous counter using a T flip-flop tracks the packet count. The counting process stops once the number of transmitted packets reaches the defined maximum packet value.



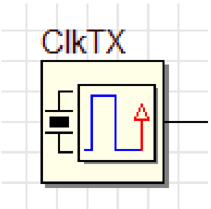
5.6 7-Segment Display

The display is used to show packet data and counter values during transmission. It has a packet data display that shows the selected packet before transmission and a counter value display that tracks and updates the number of transmitted packets. It will stop updating once the transmission is complete and all packets are sent. This enables users to monitor data selection and transmission status in real-time.



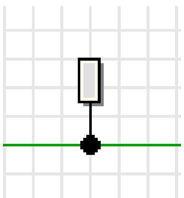
5.7 Clock Generator

The usage of the clock generator provides a continuous clock signal to synchronize clock operations. It ensures the count-up counter, flip-flops, and other sequential components toggle correctly, enabling proper packet transmission timing.



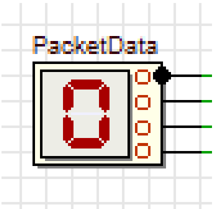
5.8 Test LED

This Test LED is used as an indicator to show the status of different components. It helps visualize key functions such as the power ON status and clock toggling which a blinking LED confirms the clock signal. Besides, it also indicates the data transmission process; LEDs light up based on active components.



5.9 Input Hex Digit

We used the input hex digit to enter the packet data for transmission. It lets users define the number of packets to be sent. This input is fed into the comparator, ensuring the correct number of packets is transmitted and monitored throughout the process.



6.0 Conclusion

To summarize our project, it successfully demonstrates the design and implementation of a packet transmission system using digital logic components. We were able to create a simulation for transferring data between two computers in different labs via a single cable by utilizing multiplexers, demultiplexers, counters, comparators, clock enablers, and monitoring systems. This project is a valuable hands-on experience as we create the circuit by using Deeds Software while applying digital logic concepts emphasizing the importance of precise timing, synchronization, and data monitoring in network communication systems. Not just that, the integration of a visual display using a seven-segment display enhanced the interactivity and usability of the system.

6.1 Reflection

Reflecting on this project, we are proud that we are able to complete the designing and implementing a functional packet transmission system using digital logic components. One of the achievements was we were able to integrate components like multiplexers, counters, and comparators to ensure the data transmission is smooth and accurate.

Our team's strength is that each and everyone contributes fresh ideas in designing and implementing the circuit. This makes our project done on-time and our project is the result of a lot of viewpoints that all team members have agreed upon.

However, we had some trouble while working on entering packet data as it is not giving the expected outcome. It was a challenge for us to keep syncing all the components smoothly.

In the future, we feel like we could work on a faster and more efficient counter and improvise the input system so that it is more user-friendly. We are also thinking about adding a password on our circuit by using a decoder.

References

1. Digital Logic Lab Manual (2024), School of Computing, Faculty of Engineering, UTMJB
2. Floyd, T.L., (2014), “Digital Fundamentals”, 10th Edition, Prentice Hall, USA.

Appendices

Nur Athirah Binti Abasa	- Designing full circuit using Digital Electronic Simulator - Introduction - Problem background - The Requirement
Afina Soleha Batrisyia Binti Mohd Hisafudin	- Designing full circuit using Digital Electronic Simulator - Suggested solution
Nur Alia Aishah Binti Nazri Sakri	- Designing full circuit using Digital Electronic Simulator - Dedication and acknowledgment - Conclusion - Reflection - Appendices
Fakhira Anisa Binti Mohd Radzi	- Designing full circuit using Digital Electronic Simulator - System Implementation

Table 1 List of task distribution